



Project Report

Project Title: Visual Servoing Autonomous Smart Car Using
Raspberry Pi 5 and Computer Vision

Course Title: CSE331

Course Name: Microprocessor Interfacing
Section: 04

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Visual Servoing Autonomous Smart Car Using Raspberry Pi 5 and Computer Vision

Project Overview

This project focuses on designing and implementing an autonomous smart robotic car capable of visually tracking and following a target using computer vision techniques. The main objective of this system is to create an intelligent mobile robotic platform that can detect a visual target, estimate its position relative to the camera, and automatically control its movement in real time.

The system is built around a Raspberry Pi 5 as the main processing unit, integrated with a USB camera for vision sensing, a motor driver for movement control, and a four-wheel robotic chassis for mobility.

Initially, the system was designed using simple image-based target following. Later, the system was improved using camera calibration, ArUco marker pose estimation, and PWM-based motor speed control to achieve more accurate and stable movement.

This project demonstrates the integration of robotics, embedded systems, computer vision, and autonomous control in a practical real-world application.

Project Objective

The primary objectives of this project were:

- To design a fully functional autonomous robotic car.
- To implement computer vision-based target tracking.
- To estimate target position using camera feedback.
- To control motor movement automatically based on visual input.
- To improve tracking accuracy using camera calibration.
- To create a modular platform for future upgrades such as object tracking, human following, and intelligent navigation.

Basic Idea of the Project

The core idea of this project was to develop a robotic vehicle that can “see” a target using a camera and automatically move toward it.

Instead of using traditional line sensors or ultrasonic sensors as the main guidance mechanism, the system uses computer vision.

The workflow of the system is:

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graph TD; A[Camera captures video] --> B[Computer vision detects target]; B --> C[Target position is calculated]; C --> D[Raspberry Pi processes movement decision]; D --> E[Motor driver receives command]; E --> F[Car moves automatically];
```

At the beginning, the system followed an ArUco marker using simple image center alignment. Later, the system was upgraded to use real-world pose estimation using camera calibration.

Hardware Components Used

The following hardware components were used in this project:

Component	Quantity	Purpose
Raspberry Pi 5	1	Main processing unit
L298N Motor Driver	1	Motor control
USB Camera	1	Visual input
4WD Robot Chassis Kit	1	Mechanical platform
BO DC Motors	4	Wheel movement

18650 Li-ion Battery	2	Motor power
Battery Holder (2S)	1	Battery connection
Power Bank	1	Raspberry Pi power
Jumper Wires	Multiple	Electrical connections
MicroSD Card	1	Operating system
USB Type-C Cable	1	Pi power supply

Software and Libraries Used

The following software tools and libraries were used:

- Raspberry Pi OS
- Python
- OpenCV
- gpiozero
- SSH for remote access
- SCP for file transfer

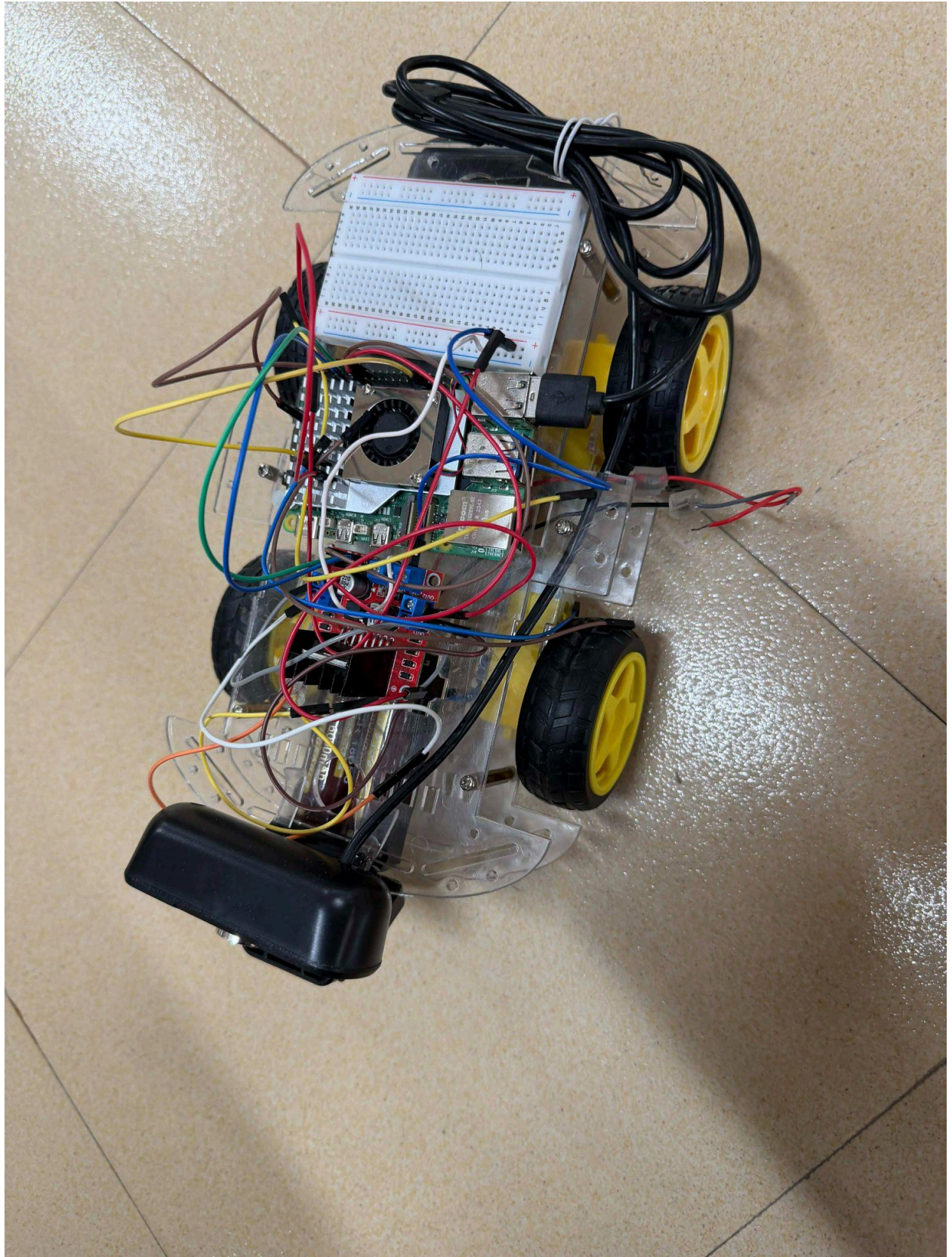
Development Process

Phase 1: Hardware Assembly

The first step was assembling the 4WD robotic chassis.

Tasks completed:

- Installed four motors
- Mounted wheels
- Mounted motor driver
- Mounted Raspberry Pi

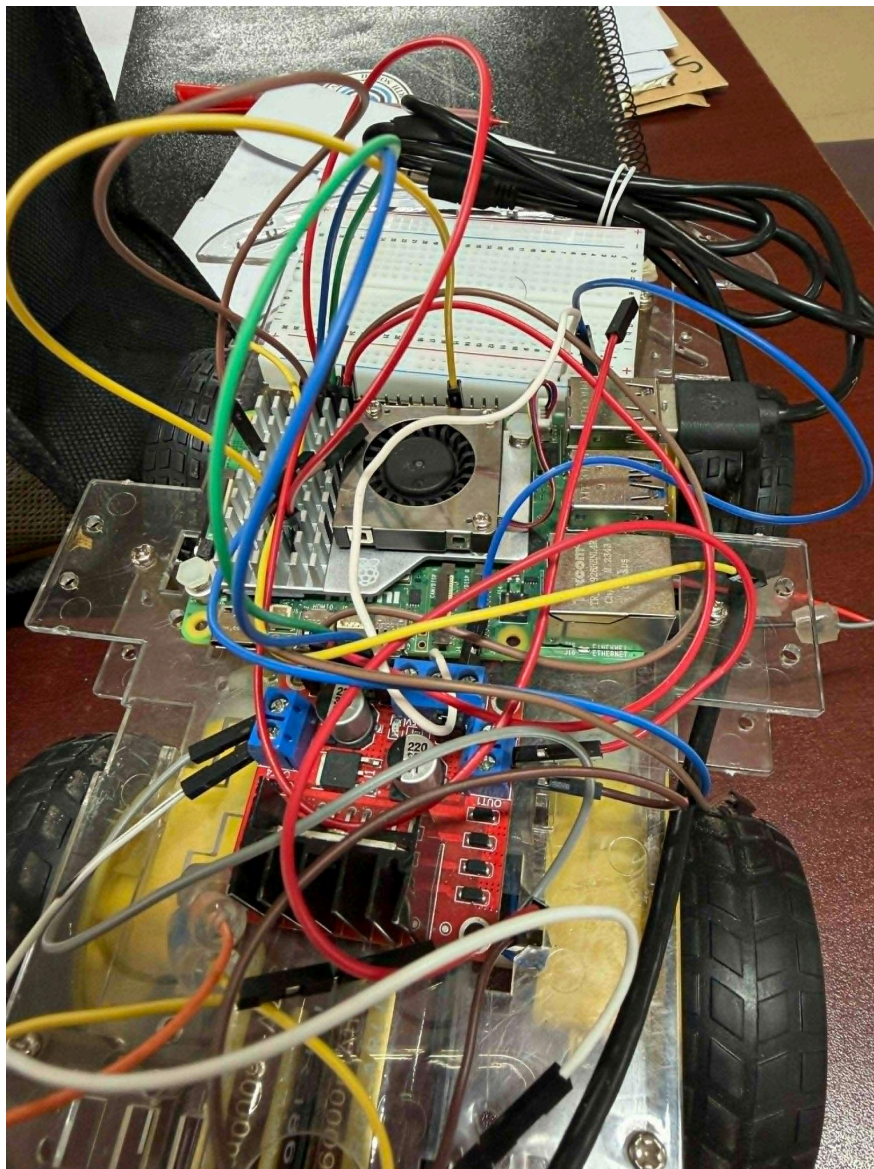


Phase 2: Raspberry Pi Setup

The Raspberry Pi was configured with:

- Operating system installation
- WiFi setup
- SSH access
- Python virtual environment setup

Initially, the Raspberry Pi was accessed remotely from a MacBook for development and debugging.



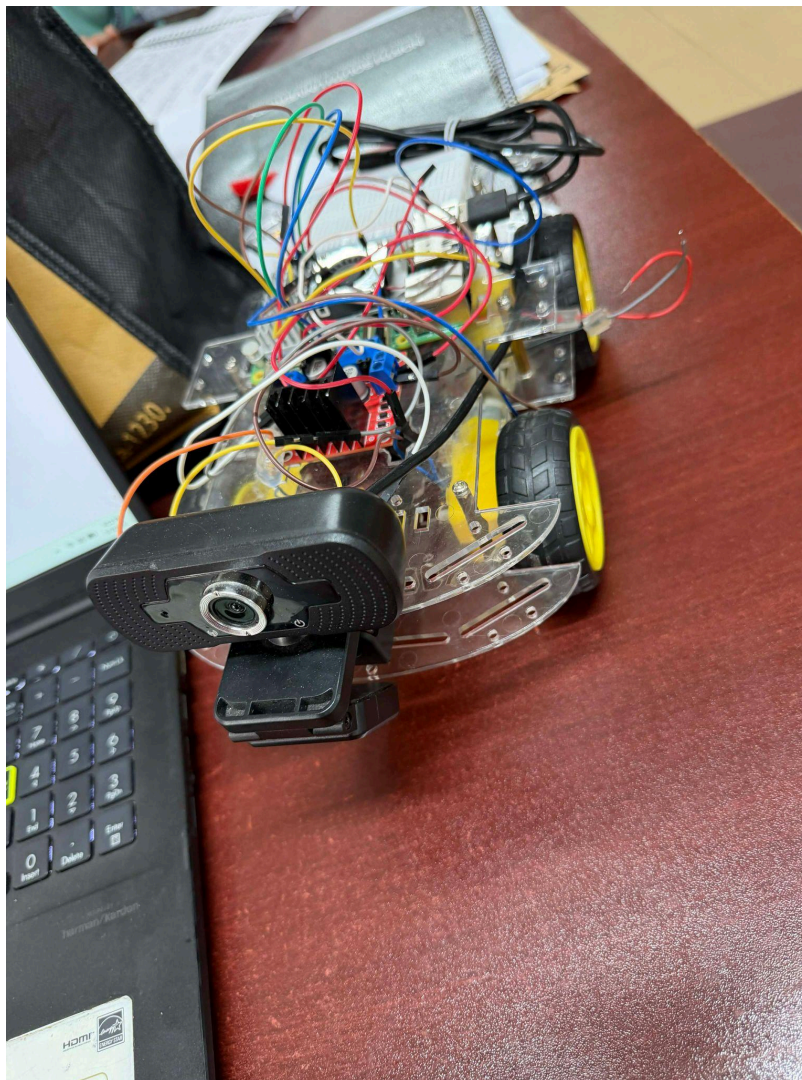
Phase 3: Camera Integration

A USB camera was connected to the Raspberry Pi.

Tasks completed:

- Camera device detection
- Image capture testing
- OpenCV camera stream testing

After successful testing, the camera was integrated into the main vision pipeline.



Phase 4: Basic Vision Tracking

The first vision algorithm used ArUco marker detection.

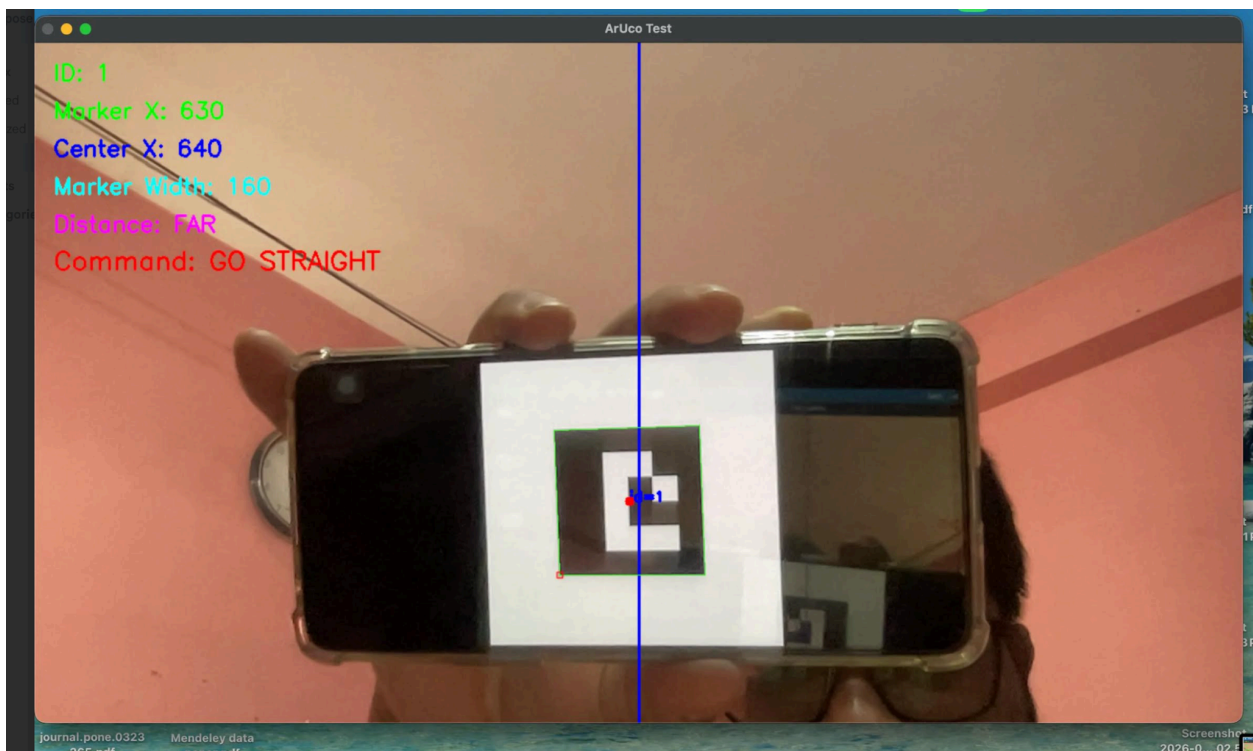
The system:

- Captured camera frames
- Detected marker ID
- Found marker center
- Compared marker center with image center
- Controlled the car based on marker alignment

Movement logic:

- Marker on left → Turn left
- Marker on right → Turn right
- Marker centered → Move forward
- Marker too close → Stop

T



his version worked successfully

Phase 5: Camera Calibration

To improve tracking accuracy, camera calibration was introduced.

Calibration process:

- Printed chessboard pattern
- Captured multiple images
- Detected chessboard corners
- Computed camera matrix
- Computed distortion coefficients

This allowed accurate 3D pose estimation of the marker.

Outputs obtained:

- X axis position
- Y axis position
- Z axis distance

This significantly improved tracking precision.



Phase 6: PWM Speed Control

The motor system was upgraded using PWM speed control.

Previously:

- Motors were either fully ON or OFF

After upgrade:

- Variable speed control was introduced
- Smooth turning became possible
- Better tracking stability was achieved

ENA and ENB pins of the motor driver were connected to Raspberry Pi PWM GPIO pins.

8. Problems Faced During Development

Several challenges were encountered during implementation.

Problem 1: Raspberry Pi WiFi Connectivity Issue

Initially, the Raspberry Pi frequently disconnected from WiFi.

Cause:

- Weak wireless signal
- Network configuration instability

Solution:

- Reconfigured WiFi settings
- Reconnected through SSH
- Verified hostname resolution

This restored stable communication.

Problem 2: Motor Driver Power Issue

Initially, motors did not move properly.

Cause:

The Raspberry Pi cannot directly supply enough current for four DC motors.

Solution:

A separate battery system was introduced:

- Two 18650 Li-ion batteries in series (2S)
- Approximate output: 6V–7.4V

This provided sufficient current for motor operation.



Problem 3: Raspberry Pi Power Issue

Initially, the Raspberry Pi showed unstable behavior.

Symptoms:

- Random disconnections
- Voltage warnings
- Camera instability

Cause:

Insufficient power supply current.

Solution:

Two changes were made:

1. Used a power bank as dedicated power source
2. Replaced USB cable with higher current output cable (3A–5A)

After this, the Pi operated stably.

Problem 4: GPIO Control Issue

GPIO libraries initially failed.

Cause:

Pin factory setup issue.

Solution:

Configured GPIO using:

```
GPIOZERO_PIN_FACTORY=lgpio
```

This solved GPIO access problems.

9. Current Project Status

Currently completed:

Chassis assembly
Raspberry Pi setup
USB camera integration
ArUco marker tracking
Camera calibration
Pose estimation
PWM motor speed control
Autonomous marker following

The car can now:

- Detect the marker
- Estimate its position
- Move left/right/forward automatically
- Stop when too close

10. Approximate Cost Analysis

Item	Approximate Cost (BDT)
Raspberry Pi 5 (8GB)	25,000
USB Camera	2,000
L298N Motor Driver	350
4WD Chassis Kit	1,500
18650 Batteries	600
Battery Holder	150
Power Bank	1,500
Jumper Wires	200
Memory Card	800

Miscellaneous 500

Total Approximate Cost:

~27,000 BDT

11. Current Limitations

Although the project is functional, some limitations remain:

- Currently tracks only ArUco markers
- No obstacle avoidance yet
- No human detection yet
- No object recognition yet
- Performance may vary under poor lighting
- Battery runtime is limited

12. Future Improvements

Future upgrades planned:

1. ORB-Based Feature Tracking

Use Oriented FAST and Rotated BRIEF to follow specific objects.

2. Human Following System

Implement human detection using:

- Face detection
- Body detection

3. Obstacle Avoidance

Add ultrasonic sensors for safe navigation.

4. AI-Based Object Detection

Integrate deep learning models for smarter tracking.

13. Conclusion

This project successfully demonstrates the implementation of an autonomous visual servoing robotic car using Raspberry Pi and computer vision.

Starting from a basic marker-following system, the project was gradually improved using camera calibration, pose estimation, and PWM motor speed control.



